

Tianyi Wang

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Research Interests: Smart Sensors, MEMS, Micro and Nano Fabrication Technologies

EDUCATION

East China University of Science and Technology (ECUST)

♦ **Major:** Process Equipment and Control Engineering | **GPA:** 90.08/100 | **Ranking:** 1/127

University of Windsor (UWindsor)

♦ **Research Assistant** (Mitacs Globalink Research Internship | **Full Scholarship**)

Shanghai, China

Sept. 2022-Present

Windsor, ON, Canada

July 2025-Oct. 2025

PUBLICATIONS

- [1] Zhang, C., **Wang, T.**, Zhang, G., Gao, R., Gao, C., Wang, Z., & Xuan, F. (2025). Rational design and fabrication of MEMS gas sensors with long-term stability: A comprehensive review. Advanced Science, e11555. (doi.org/10.1002/adv.202511555)
- [2] **Tianyi Wang**, Longqi Wu, Guozhu Zhang*, et al. (2025). Humidity-Tolerant Chemiresistive Hydrogel Sensor for Real-Time Breath CO₂ Monitoring. (**Major Revision, Chemical Engineering Journal**)
- [3] Chao Zhang, **Tianyi Wang**, Guozhu Zhang*, et al. (2025). S-TCN Neural Networks for Gas Detection: Algorithm Design and Implementation on Edge Devices. (Submitted)
- [4] **Tianyi Wang**, Guozhu Zhang, et al. (2025). Hydrogel-Integrated Carbon Dioxide Sensor, Preparation Method Thereof, and Smart Mask. CN 2025111826102 (Patent)
- [5] **Tianyi Wang**, Guozhu Zhang, et al. (2025). Ion-Sensitive Field-Effect Transistor (ISFET) Sensor. CN 2025218665554 (Patent)
- [6] Guozhu Zhang, Chengze Gao, **Tianyi Wang**, et al. (2024). Manufacturing Methods and Application of CO₂ Gas Sensor Based on Hydrogel Structure. CN 2024110896635 (Patent)

RESEARCH EXPERIENCE

Embedded Hardware Development for GroGenesis V2 Greenhouse Sensor System

Research Assistant (Mitacs Globalink Research Internship)

July 2025-Present

- Participated in circuit design and multilayer PCB layout for a miniaturized environmental monitoring node.
- Performed soldering, debugging, and functional testing on multiple hardware prototypes to ensure stable sensor communication and power regulation.
- Developed Arduino-based firmware and implemented long-term Wi-Fi data logging and stability monitoring to support cloud acquisition and remote visualization.

S-TCN Neural Networks for Gas Detection: Algorithm Design and Implementation on Edge Devices

Key Contributor

June 2024-July 2025

- Built a multi-channel gas sensor array and designed a 4-layer PCB.
- Developed an S-TCN model using dilated depthwise separable convolutions to improve accuracy and efficiency.
- Deployed the model on a microcontroller, achieving real-time gas identification with over 98% accuracy.

Rational Design and Fabrication of MEMS Gas Sensors toward Long-Term Stability: A Comprehensive Review

Key Contributor

Sept. 2024-Apr. 2025

- Reviewed MEMS gas sensors and identified core failure modes affecting long-term operational stability.
- Analyzed material degradation mechanisms and evaluated stable electrode/sensing materials through literature mining and theoretical comparison.

Humidity-Tolerant Chemiresistive Hydrogel Sensor for Real-Time Breath CO₂ Monitoring

First Author & Leader

July 2024-Mar. 2025

- Synthesized and optimized dimethylamine-functionalized hydrogels with enhanced CO₂ sensitivity, mechanical stretchability, and humidity-assisted ionic conductivity.
- Designed a flexible PCB with integrated microcontroller, analog interface, and wireless modules for real-time monitoring.
- Integrated the sensor into a wearable smart mask and developed a mobile interface for real-time monitoring.

Design and Development of the Ion-Sensitive Field-Effect Transistor (ISFET) PH Sensor

Leader

Sept. 2023-Dec. 2024

- Designed an epitaxial-gate ISFET sensor with a separated chip-packaging architecture to enhance long-term stability, replaceability, and signal isolation.
- Developed a compact signal acquisition circuit integrating low-noise amplification, temperature compensation, and reference electrode interfacing for stable pH readout.

HONOUR & AWARDS

First Prize, International Innovation and Entrepreneurship Competition	2025
First Prize, Challenge Cup National Competition	2025
National Scholarship (First in Major) & Top-tier School Scholarship (Top 3%)	2024
Yihai Kerry National Scholarship	2024
ECUST Outstanding Student (Top 6%)	2024
ECUST First-Class Student (Top 6%)	2023

SKILLS

- Techniques:** Altium Designer, Ansys, MATLAB, 3ds Max, CAD, SolidWorks, Python, C
- Languages:** Mandarin Chinese (Native), English (TOEFL: 104 / Best Score:106)